

Taxi Cancellation Prediction

Final Project Report

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Business Analytics Problem: Predicting whether taxi bookings are likely to be cancelled using R-based data mining methods.

Executive Summary

This final project analyzes a taxi cancellation prediction problem using the dataset Taxi-cancellation-case.csv and R. The business goal is to identify taxi bookings that are more likely to be cancelled so a taxi company can improve driver allocation, reduce wasted trips, and support better operational planning. The dataset contains 10,000 booking records and 19 original variables, including vehicle model, package type, travel type, pickup and drop-off areas, booking channel, dates, geographic coordinates, and the target variable Car_Cancellation.

The main analytical finding is that the data is highly imbalanced: 92.57% of bookings were not cancelled and only 7.43% were cancelled. Because of this imbalance, overall accuracy can be misleading. A model can achieve high accuracy by predicting most bookings as non-cancellations, even if it fails to identify the cancellation cases that matter most to the business.

Three major methods were emphasized: K-Nearest Neighbors, Decision Tree, and Boosting. Logistic Regression was also attempted as a baseline model, but it could not produce valid predictions because the validation data contained a vehicle_model_id level not present in the training set. Among the working models, Boosting performed best overall. It had the highest accuracy at 92.32%, the strongest sensitivity at 12.06%, and the best F1 score at 0.1984. Although its sensitivity was still limited, Boosting was the only model that meaningfully improved cancellation detection compared with KNN and Decision Tree.

1. Description of the Business Problem

Taxi cancellation is an operational problem. When customers cancel bookings, taxi companies may experience wasted driver time, inefficient vehicle assignment, lost revenue, and service disruptions. If the company can predict which bookings are more likely to be cancelled, it can make better decisions about driver dispatching, customer follow-up, booking confirmation, and resource planning.

The business analytics question for this project is:

Can data mining models built in R predict taxi booking cancellations well enough to support better operational decisions?

The practical challenge is not only predicting the majority outcome. Since most bookings are not cancelled, the model must be evaluated based on its ability to detect the minority class: actual cancellations. Therefore, this project uses accuracy, sensitivity, specificity, precision, F1 score, confusion matrices, and gains charts to evaluate model performance.

2. Description of the Data

The dataset contains 10,000 observations and 19 original variables. Each row represents a taxi booking record. The target variable is Car_Cancellation, where 0 means the booking was not cancelled and 1 means the booking was cancelled.

Data Component	Description
Number of records	10,000 taxi booking observations
Original variables	19 variables
Target variable	Car_Cancellation: 0 = not cancelled, 1 = cancelled

Main predictors	Vehicle model, package type, travel type, pickup/drop-off areas, city IDs, online booking, mobile booking, date/time features, and trip length
Training/validation split	60% training data and 40% validation data
Training records	6,000 records
Validation records	4,000 records

Class imbalance was the most important characteristic of the dataset. Out of 10,000 bookings, 9,257 were not cancelled and 743 were cancelled. This means only 7.43% of the dataset belongs to the cancellation class.

Class	Frequency	Percentage
0 = Not Cancelled	9,257	92.57%
1 = Cancelled	743	7.43%

3. Data Preparation and Feature Engineering

The data preparation process was completed in R before model training. The main preparation steps were:

- Missing values were replaced with 0 so the models could run without NA-related errors.
- Categorical ID variables such as `vehicle_model_id`, `package_id`, and `travel_type_id` were converted into factors.
- Date fields were converted into usable time-based features, including pickup day of week, pickup hour, and booking-created day of week.
- Trip length was calculated from latitude and longitude coordinates using a distance function.
- The dataset was split into a 60% training set and a 40% validation set using `set.seed(1)` for reproducibility.

The engineered features helped translate raw booking information into variables that could be used by the models. For example, pickup hour and day of week may capture timing patterns, while trip length may reflect differences in cancellation behavior across shorter and longer trips.

4. Analytic and Data Mining Techniques Used

The course requirement asks for 2 to 3 analytic or data mining techniques. This project discusses K-Nearest Neighbors, Decision Tree, and Boosting as the main techniques. Logistic Regression was also attempted as a baseline, but it was not included as a final working model because it failed during prediction due to factor-level issues.

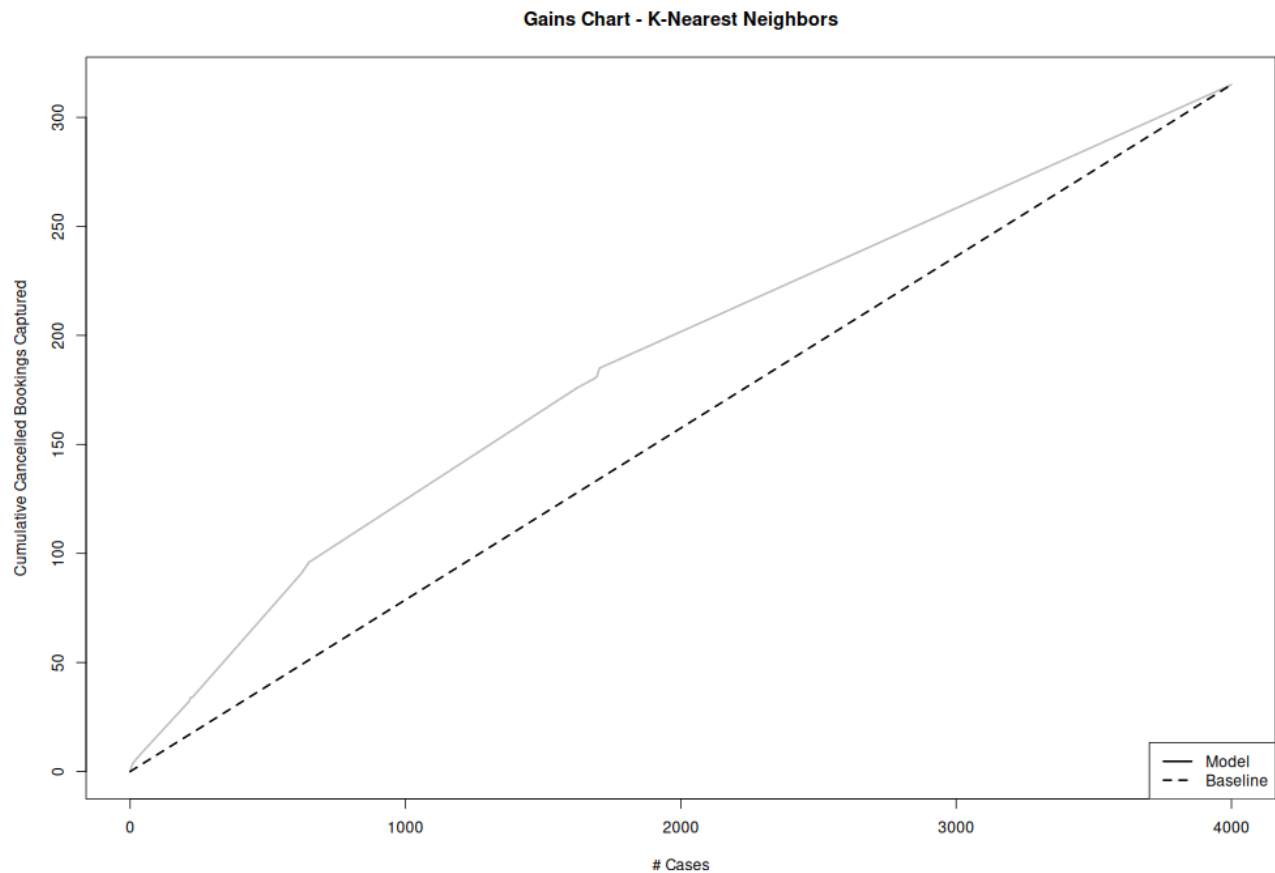
4.1 K-Nearest Neighbors

K-Nearest Neighbors (KNN) is a classification method that predicts a case based on the classes of nearby cases. In this project, KNN was trained using the `caret` package and tuned using $k = 1$ and $k = 10$. The final selected value was $k = 10$ because it produced the highest resampling accuracy.

Metric	KNN Result
Accuracy	0.9212
Sensitivity	0.0063
Specificity	0.9995
Precision	0.5000

F1 Score	0.0125
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Interpretation: KNN achieved high accuracy, but its sensitivity was extremely low. It only correctly identified 2 cancellations out of 315 actual cancellations in the validation data. This means KNN mostly predicted the majority class and was not very useful for detecting cancelled bookings.



4.2 Decision Tree

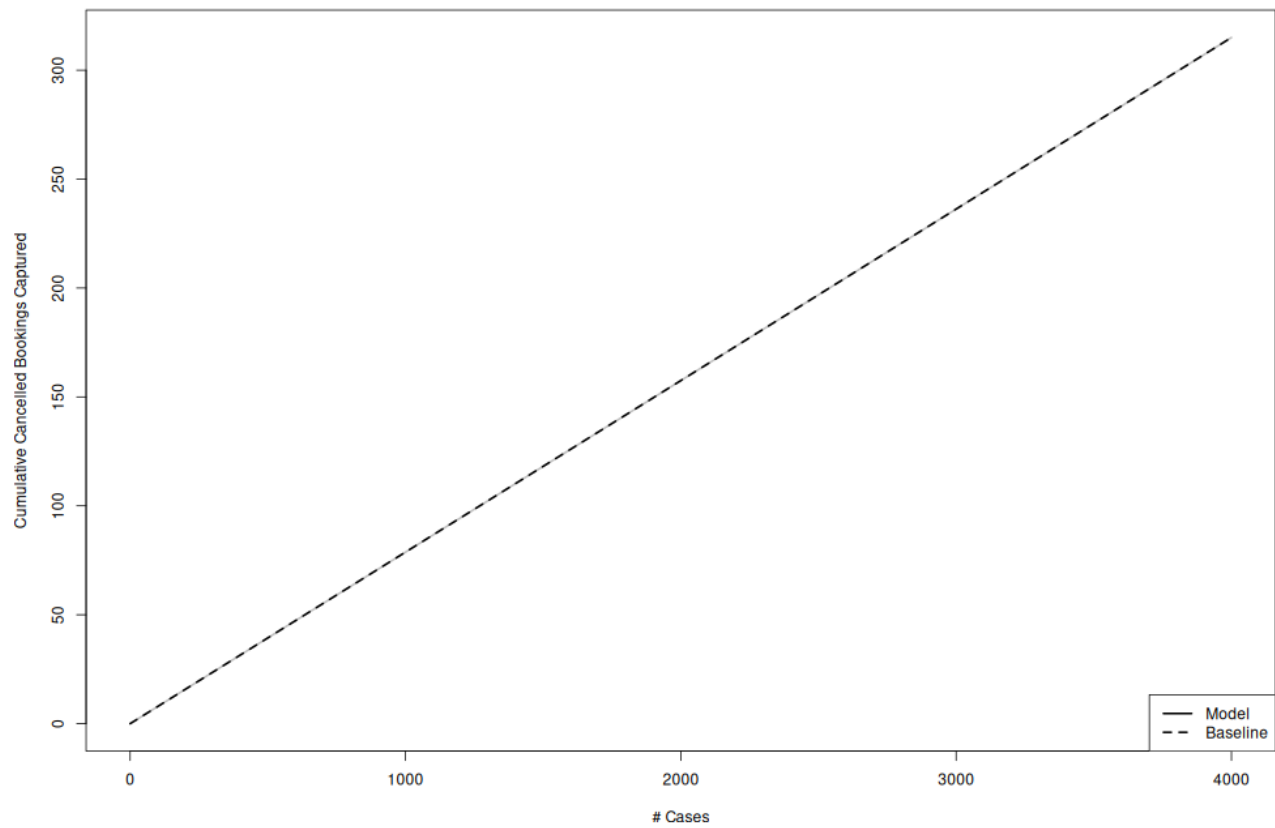
A Decision Tree is an interpretable classification method that creates decision rules by splitting the data based on predictor variables. In this project, a tree was built using rpart and then pruned using the complexity parameter selected by cross-validation error.

The pruned tree collapsed into a single-node tree. This means the model did not create any useful decision splits and predicted the majority class, non-cancellation, for the full dataset. The single node showed approximately 93% non-cancellations and 7% cancellations, which directly reflects the class imbalance in the data.

Metric	Decision Tree Result
Accuracy	0.9212
Sensitivity	0.0000
Specificity	1.0000
Precision	NA
F1 Score	NA

Interpretation: The Decision Tree had the same accuracy as the no-information rate, but it detected zero actual cancellations. This shows that high accuracy was misleading because the model simply predicted non-cancellation for every validation case.

Gains Chart - Decision Tree



Pruned Decision Tree - Taxi Cancellation Prediction

0
.93 .07
100%

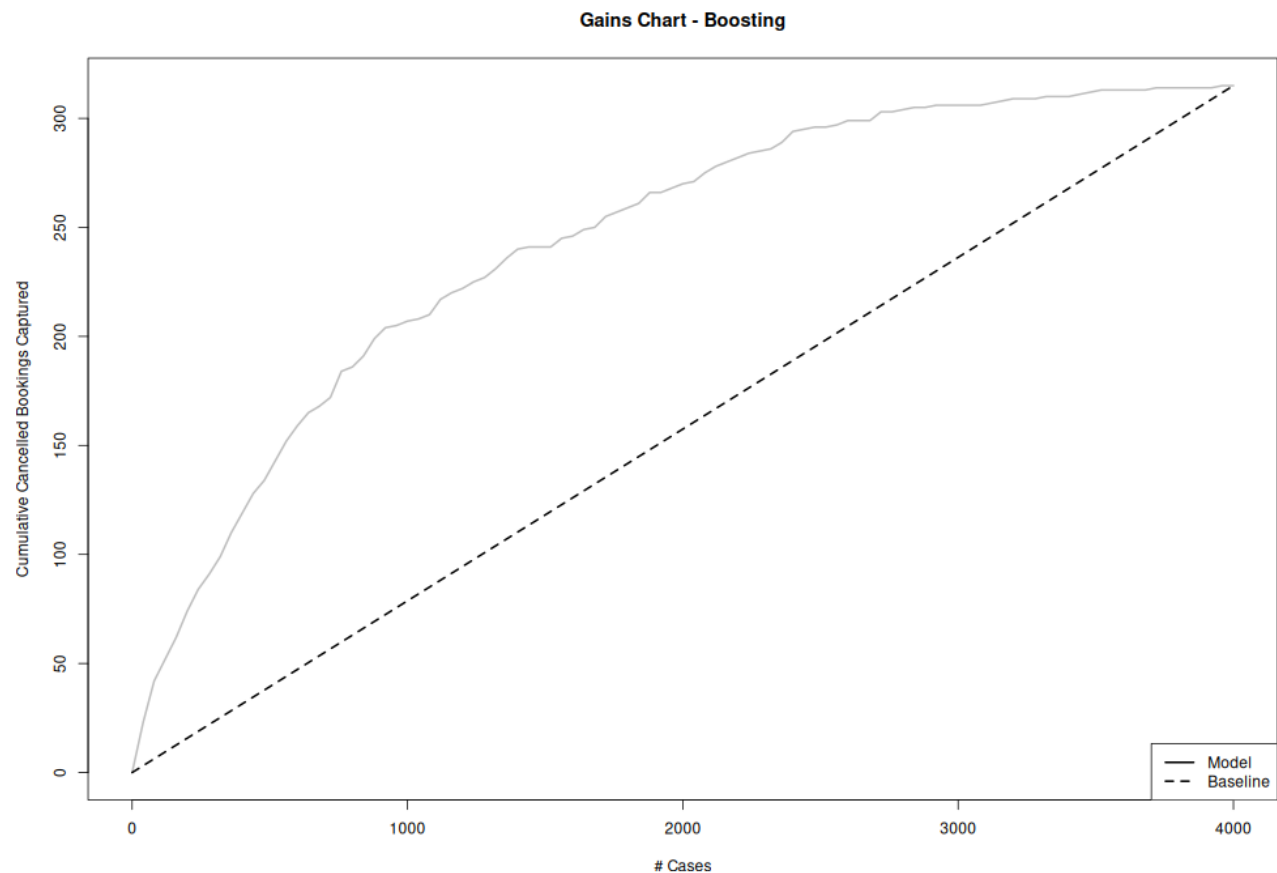
4.3 Boosting

Boosting is an ensemble method that combines multiple weak learners to improve prediction performance. In this project, Boosting was implemented with the adabag package. It was the strongest model because it improved cancellation detection while maintaining high overall accuracy.

Metric	Boosting Result
Accuracy	0.9232
Sensitivity	0.1206
Specificity	0.9919
Precision	0.5588
F1 Score	0.1984

Interpretation: Boosting correctly identified 38 of the 315 actual cancellations in the validation data. This sensitivity is still limited, but it is much better than KNN and Decision Tree. Its gains chart also showed a clear improvement over the baseline, especially in the highest-risk cases.

Boosting variable importance: The most important predictors included trip_length, vehicle_model_id, from_area_id, booking_created_DOW, to_area_id, from_date_Hour, and online_booking. This suggests that cancellation behavior may be related to trip distance, booking channel, timing, and pickup/drop-off location patterns.



4.4 Logistic Regression as a Baseline Attempt

Logistic Regression was attempted as a traditional baseline classification model. However, the model could not produce valid validation predictions because the validation dataset contained a `vehicle_model_id` level that was not present in the training data. The error was: `factor vehicle_model_id has new levels 36`.

This result is still useful as a limitation. It shows that categorical predictors with many levels can create instability when some categories appear in validation data but not training data. For future work, this issue could be handled by combining rare categories, using consistent factor levels before splitting the data, or using modeling methods that handle unseen categories more robustly.

5. Discussion of Results

The final model comparison shows why accuracy cannot be the only evaluation metric. KNN and Decision Tree both achieved 92.12% accuracy, but they had extremely poor sensitivity. KNN detected only 0.63% of cancellations, while the Decision Tree detected none. This happened because the dataset is highly imbalanced and most records are non-cancellations.

Model	Accuracy	Sensitivity	Specificity	Precision	F1
K-Nearest Neighbors	0.9212	0.0063	0.9995	0.5000	0.0125
Decision Tree	0.9212	0.0000	1.0000	NA	NA
Boosting	0.9232	0.1206	0.9919	0.5588	0.1984
Logistic Regression	NA	NA	NA	NA	NA

Boosting was the best overall model. It achieved the highest accuracy, the highest sensitivity, the strongest F1 score, and the clearest improvement over the baseline in the gains chart. However, the sensitivity of 12.06% also shows that the model still missed many cancellations. Therefore, the model is useful as a starting point for identifying higher-risk bookings, but it should not be treated as a perfect cancellation prediction system.

6. Implications of the Findings

The findings have several practical implications for a taxi company:

- Predictive modeling can help identify higher-risk bookings, especially when using Boosting rather than simpler models.
- The company should not rely only on accuracy because the dataset is imbalanced and accuracy favors the non-cancellation class.
- A gains chart is useful because it shows whether a model can rank likely cancellations above the baseline.
- Operational teams could prioritize confirmation calls, driver assignment caution, or additional verification for bookings ranked as high risk by the model.
- The strongest predictors suggest that cancellation risk may be connected to trip length, vehicle model, pickup/drop-off location, booking timing, and booking channel.

The most important business recommendation is to use Boosting as the preferred model among those tested, while continuing to improve the model with better imbalance-handling techniques. For

example, future versions could test resampling methods, different probability thresholds, cost-sensitive learning, or additional predictors to improve cancellation detection.

7. Limitations

- The dataset is highly imbalanced, with only 7.43% cancellations.
- Replacing all missing values with 0 may introduce bias because missingness can have different meanings depending on the variable.
- The Decision Tree collapsed into a single-node model, showing limited predictive value after pruning.
- Logistic Regression failed during prediction because the validation data contained a `vehicle_model_id` level not seen in training.
- Boosting improved cancellation detection, but its sensitivity was still only 12.06%, so many cancellations remained undetected.
- Trip length values included very large distances, suggesting the need for possible outlier investigation in future work.

8. Conclusion

This project applied data mining methods in R to analyze a taxi cancellation prediction problem. The dataset contained 10,000 booking records and a strongly imbalanced target variable, with only 7.43% cancellations. This imbalance shaped the entire modeling problem and made accuracy alone an unreliable evaluation metric.

The main conclusion is that Boosting was the strongest model among the methods tested. KNN and Decision Tree both produced high accuracy, but they did not meaningfully detect cancellations. The Decision Tree was especially limited because the pruned tree became a single node that predicted the majority class. Boosting was better because it achieved the highest accuracy and the best sensitivity and F1 score, while also performing above the baseline in the gains chart.

From a business perspective, this project shows that predictive analytics can support taxi operations, but model evaluation must focus on the actual business objective: detecting likely cancellations. A company should use models like Boosting to rank bookings by cancellation risk, but should continue improving the model before using it for high-stakes automated decisions.